As before, the coefficients γ and D are given by double Fourier sine and cosine transforms, respectively (see the discussion in Sect. 5.2.3 for details). Thus we can readily write down the resulting expressions [compare (5.46) and (5.48)],

$$\gamma = \frac{\pi}{2}\frac{1}{M\Omega}J_{\mathrm{eff}}(\Omega,T) = \gamma_0\frac{\Lambda^2}{\Lambda^2+\Omega^2}\tanh\left(\frac{\Omega}{2k_{\mathrm{B}}T}\right), \tag{5.200}$$

$$D = \frac{\pi}{2}J_{\mathrm{eff}}(\Omega,T)\coth\left(\frac{\Omega}{2k_{\mathrm{B}}T}\right) = M\gamma_0\Omega\frac{\Lambda^2}{\Lambda^2+\Omega^2}. \tag{5.201}$$